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SPECIAL REPORT

Management of antithrombotic therapies in kidney transplantation candidates and recipients: an ERBP clinical practice document by the DESCaRTES Working Group of ERA and the EKITA Committee of ESOT

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ABSTRACT

Thromboembolic and atherosclerotic disorders, necessitating antithrombotic therapies, are prevalent among patients with advanced chronic kidney disease (CKD), as well as in kidney transplant (KT) recipients. However, the optimal management of antithrombotic therapies in these patients remains uncertain due to conflicting evidence and the exclusion of advanced CKD and KT recipients from major randomized controlled trials. Additionally, the perioperative management of antithrombotic therapy in end-stage kidney disease patients undergoing transplantation requires specific attention. This clinical practice document prepared by experts from the DESCaRTES Working Group of the European Renal Association (ERA) and European Kidney Transplant Association (EKITA) from European Society of Organ Transplantation (ESOT) following European Renal Best Practice (ERBP) guidelines is designed to support clinicians in decision making on antithrombotic therapies in KT candidates and recipients. Following a detailed systematic literature review, it presents current knowledge in epidemiology and pathophysiology and summarizes all relevant studies in this topic.

Keywords: anticoagulation, antiplatelet agents, antithrombotic therapy, kidney transplant candidates, kidney transplant recipients

INTRODUCTION

Patients with end-stage kidney disease (ESKD), especially those requiring dialysis, are at high risk for cardiovascular events, including myocardial infarction, stroke, and other thromboembolic events such as deep vein thrombosis and pulmonary embolism [1]. Despite uncertainties regarding the efficacy and safety of antiplatelet or anticoagulant drugs in patients with chronic kidney disease (CKD) stage 4–5, many of these patients receive antithrombotic therapies [2]. Even though the exten-

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sive transplant evaluation process sometimes excludes ESKD patients with severe comorbid conditions requiring antithrombotic therapies from waitlisting, currently the use of antiplatelet or anticoagulant drugs has not been accepted as a contraindication for kidney transplantation. Indeed, recent KDIGO guidelines for kidney transplant (KT) candidates recommend that ESKD patients should not be excluded from consideration for transplantation due to their use of antiplatelets or anticoagulants *per se* [3]. Nevertheless, due to broad prescription of antithrombotic drugs, the perioperative and early post-transplant management of hemostasis has become increasingly difficult.

Additionally, in the post-transplant period antiplatelet or anticoagulant drugs are often prescribed. In patients with atrial fibrillation on dialysis, anticoagulant drugs may be (re)started after transplantation. Moreover, rates of newly diagnosed cerebrovascular events and myocardial infarction, requiring antithrombotic therapies, are reported up to 6.8% and 11.1%, respectively, within 3 years post-KT [4–6]. The incidence of major cardiovascular adverse events (defined as myocardial infarction, stroke, critical peripheral ischemia and instable angina) occurs in 34% of living donor KT recipients within five years according to a Swedish study [7]. In parallel, almost 9% of KT recipients developed venous thromboembolism (VTE) within 5 years posttransplant, which is a seven-fold higher risk as compared with the general population [8]. These elevated risks underscore a need for guidance on antithrombotic therapy dosing, drug–drug interactions, and tailored strategies for high-bleeding-risk procedures like allograft biopsies. Yet, the evidence base remains limited, as most large clinical trials assessing antithrombotic efficacy and safety have excluded dialysis patients and KT recipients, leaving a critical gap in robust, transplant-specific recommendations.

This clinical practice document aims to help joint decision-making around management of antithrombotic therapies in KT candidates and recipients. The document summarizes current knowledge of the pathophysiology of increased bleeding and thrombosis risk in ESKD patients who are KT candidates, in ESKD patients undergoing transplant surgery and KT recipients. Additionally, it provides a comprehensive summary of the current evidence on peri-operative antithrombotic therapy management in adult KT candidates, explores the available data on safety and efficacy of antithrombotic therapies in KT recipients and examines the need for interruption of antithrombotic therapy prior to allograft biopsy.

PATHOPHYSIOLOGY OF BLEEDING AND THROMBOSIS IN ESKD

ESKD has a profound influence on the hemostatic system. Alone or combined with kidney replacement therapy, it disrupts the fine balance between pro-coagulation and anticoagulation mechanisms, resulting simultaneously in both increased bleeding and thrombotic events [9]. Presently, the delineation of the patients exhibiting a predisposition toward either thrombosis or bleeding remains indeterminate.

Bleeding risk

ESKD patients have an elevated risk of clinically significant bleeding, which is primarily attributed to the uremia-associated bleeding diathesis, but may also be influenced, at least in part, by the administration of antithrombotic therapies [9]. The interplay between altered coagulation cascades, dysregulated platelet function and abnormalities in the endothelium contributes to an increased risk of bleeding events [9–11].

It is well known that diminished platelet activation and adhesion caused by uremia increases the risk of bleeding. These platelets present with decreased expression and functional impairment of glycoprotein Ib and IIb/IIIa receptors, resulting in reduced release of alpha-granules [12]. In addition, the uremic milieu alters the production of arachidonic acid and prostaglandin compounds by the platelets, therefore leading to decreased synthesis of thromboxane A2 [13, 14]. In anemic ESKD patients, additional factors amplify this risk: reduced adenosine diphosphate release, aberrant prostaglandin I2 activation and diminished platelet-vessel wall interaction, compounded by elevated nitric oxide levels that inhibit aggregation [15].

Given that ESKD patients are at high risk of cardiovascular disease, they are commonly prescribed antiplatelet drugs, thereby increasing their risk of bleeding [10]. In addition, some anticoagulants are eliminated through the kidneys and therefore require timely dose adjustments to protect against bleeding events.

Thrombosis risk

In addition to the aforementioned increased bleeding risk, ESKD patients are also paradoxically at risk of thromboembolic events [16]. This is mainly due to the activation of the renin–angiotensin–aldosterone system and endothelial cell damage accompanied by higher levels of fibrinogen and low levels of circulating fibrinolytic factors, secondary to chronic inflammation [17]. Clinically significant thrombosis in ESKD patients commonly manifests as deep venous thrombosis, pulmonary embolism or vascular access–associated thrombosis [18]. Furthermore, arterial thrombosis, frequently associated with atherosclerosis, presents as acute coronary syndrome, cerebrovascular events or peripheral artery disease [19]. It should also be noted that antiphospholipid syndrome, severe nephrotic syndrome and hereditary thrombophilic disorders can yield recurrent and severe thrombotic complications.

Effect of kidney replacement therapies on hemostasis

Although hemodialysis (HD) improves platelet function by removing uremic toxins, it may paradoxically worsen bleeding tendencies by causing platelet dysfunction secondary to the dialysis membrane components [20]. HD patients also have an increased thrombotic risk due to chronic inflammation and a multitude of procoagulant factors: contact between FXII and artificial surfaces, e.g. dialysis membranes (mainly those with negative charge), activates the intrinsic coagulation pathway, resulting in thrombin production and fibrin formation [21]. Thus, cross-striated fibrin is deposited on the dialysis membrane, leading to mechanical trapping and damage of platelets passing through, ultimately contributing to transient thrombocytopenia during the session. Additionally, platelets are directly exposed to several activating forces during HD. Contact with dialysis membranes and mechanical stress from the roller pump induces platelet degranulation and surface glycoprotein loss [22]. Furthermore, variations in blood flow within the circuit might also induce the formation of microbubbles that further promote intradialytic platelet activation [23]. Additional aspects of HD, such as biocompatibility issues or complement activation, may impair platelet function or reduce their responsiveness. Consequently, because of excessive intradialytic activation and functional impairment, platelets are frequently and uncontrollably activated during HD, a fact that leads to thrombocytopenia

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during HD sessions or post-dialysis thrombocytopathy, often termed "exhausted platelets" [23].

To reduce the risk of clotting of the dialysis filter and prevent vascular access thrombosis, heparin or unfractionated low-molecular weight heparin is commonly administered during dialysis sessions. Prolonged heparin administration in dialysis patients may cause heparin-induced thrombocytopenia (HIT) type II, which promotes platelet activation and thrombosis.

Peritoneal dialysis patients may also present with hemostatic disruptions. Studies have shown higher levels of prothrombotic proteins [apolipoprotein (a), plasminogen activator inhibitor type 1, fibrinogen, coagulation factors II, VII, VIII, IX, X, XI and XII] in peritoneal dialysis patients compared with HD patients, a fact that may contribute to a higher risk for thrombotic events [24].

PATHOPHYSIOLOGY OF BLEEDING AND THROMBOSIS DURING KIDNEY TRANSPLANTION SURGERY

Kidney transplantation surgery is associated with a high risk of bleeding, which can impair graft performance and overall patient outcomes. The incidence of severe post-transplantation bleeding has been estimated at 4%, a rate lower than that for other solid organ transplantations [25, 26]. Apart from pre-existing risk factors for bleeding, dissection of poor-quality vessels and hemodilution can also exacerbate a bleeding tendency. Achieving good surgical hemostasis is critical to reduce the risk of post-operative bleeding, prevent reoperation, blood transfusions and prolonged hospitalization. Apart from the direct effects of bleeding on graft and patient outcomes, the necessity for blood transfusion due to peri-operative bleeding may cause inadvertent human leukocyte antigen sensitization [27]. To minimize blood loss, local administration of hemostatic agents may ameliorate challenging complex bleeding during KT surgery. They are available in various forms, including gauze, mesh, solutions, sponges, powders, foams and gels, and can feature thrombin and fibrin compounds of increasing sophistication. Future developments would include nanotechnology to ensure accurate delivery [28].

Renal graft thrombosis is a feared complication of KT surgery that affects <1% of transplants [25]. Ischemia–reperfusion injury, prolonged cold ischemic time, vascular clamping and kinking of the arteries promote thrombosis through endothelial injury. Risk of thrombosis can be further increased by the presence of specific underlying diseases, such as hereditary thrombophilia and antiphospholipid syndrome [29].

A thorough preoperative evaluation of the patient's coagulation profile (platelet count, prothrombin time, activated partial thromboplastin time) and need for antithrombotic therapies are critical components of a multifaceted approach to balance bleeding and thrombosis risk. It is advisable that anticoagulation specialists be consulted before transplantation in patients with increased risk for thrombosis or bleeding, including patients with a medical history of recurrent thrombosis, antiphospholipid syndrome or hemophilia to ensure safe transplantation surgery and to support recipient recovery and well-being.

PATHOPHYSIOLOGY OF BLEEDING AND THROMBOSIS IN KT RECIPIENTS

Kidney transplantation appears to restore the balance between bleeding and thrombosis risks in most recipients. However, pre-existing diseases, use of antiplatelet or anticoagulant drugs and poor graft function can destabilize this fragile balance. Complement activation triggered by donor-specific antibodies, is often linked to antibody-mediated rejection and can contribute to the procoagulant response by activating thrombin [30]. Previous studies also showed that cytomegalovirus infection, post-transplant erythrocytosis, treatment with high-dose steroids and previous thrombotic events can increase the risk of thrombosis in KT recipients [31]. Using antithrombotic therapies, kidney allograft biopsy may be sometimes delayed due to a need for preprocedural discontinuation of the antithrombotic drugs.

Figure 1 summarizes hemostatic changes in KT candidates, patients undergoing KT surgery and KT recipients.

ANTITHROMBOTIC THERAPIES

Antiplatelet agents

Antiplatelet agents, which include acetylsalicylic acid (ASA) and $P2Y_{12}$ inhibitors like clopidogrel, ticagrelor and prasugrel, are used to prevent thromboembolic events, particularly in individuals with atherosclerotic vascular diseases, and are commonly prescribed in ESKD patients and KT recipients [32], although data on their efficacy and safety in these specific populations are lacking. ASA irreversibly inhibits cyclooxygenase-1 (COX-1), thereby impeding the synthesis and release of thromboxane A2 (TXA2), a potent platelet activator. The drug oral bioavailability typically ranges from 40% to 50%, with peak plasma concentrations usually within 30–40 min following ingestion (3–4 h for enteric-coated formulations). ASA antiplatelet effects persist for the lifespan of platelets (7–10 days). Given that around 10% of the platelet population regenerates daily, once-daily dosing is generally sufficient to maintain near-complete inhibition [32].

Clopidogrel, a second-generation thienopyridine prodrug, requires intestinal absorption and hepatic metabolism to generate an active metabolite that irreversibly binds to the $P2Y_{12}$ receptor on platelets for their entire lifespan. Kidney dysfunction markedly hampers the drug's biotransformation process, while genetic polymorphisms affecting absorption and cytochrome P450 (CYP) enzymes contribute to unpredictable variations in platelet reactivity [33]. In contrast, ticagrelor, a third-generation $P2Y_{12}$ blocker, binds extensively (approximately 99.8%) to plasma proteins without requiring metabolic activation. It achieves maximal platelet inhibition in about 2 h, with effects waning after 12 h , necessitating twice-daily dosing. A study has shown that compared with clopidogrel, ticagrelor consistently achieves more effective $P2Y_{12}$ inhibition, resulting in a lower incidence of high on-treatment platelet reactivity and drug non-responsiveness [34]. Prasugrel, another third-generation $P2Y_{12}$ blocker, carries a higher bleeding risk, especially in older adults (>75 years), low body weight (<60 kg) and those with previous stroke/transient ischemic attack [35].

Ticlopidine is a thienopyridine class antiplatelet agent that irreversibly inhibits the $P2Y_{12}$ receptor on platelets. The potential for bone marrow suppression renders it less advantageous for KT recipients who already have a high risk of depleted bone marrow due to immunosuppressive drugs.

Dipyridamole is a phosphodiesterase inhibitor that increases intracellular cAMP levels, leading to inhibition of platelet activation and vasodilation. It also inhibits adenosine uptake, enhancing vasodilatory and antithrombotic effects. There are more effective drugs for prevention of thrombotic events, so it is less favorable, but it is safe in CKD, as it does not require renal dose adjustments.

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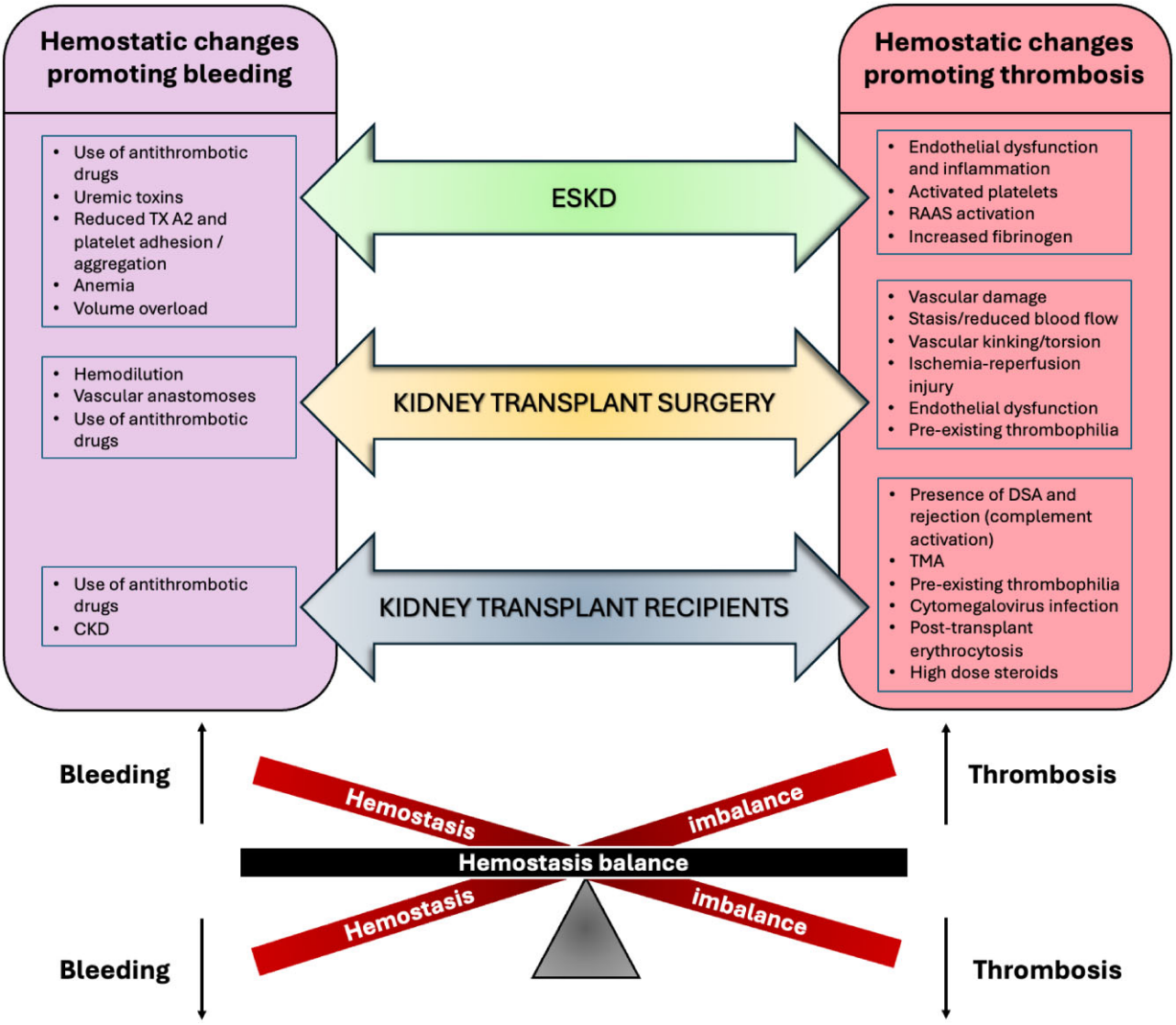

Figure 1: Hemostatic balance in ESKD, transplant surgery and KT recipients.

Table 1 summarizes the mechanisms, pharmacodynamics and KT-specific considerations for oral antiplatelet drugs.

The main indications for antiplatelets in KT recipients include: (i) secondary prevention of atherosclerotic cardiovascular disease, e.g. prior myocardial infarction, acute coronary syndrome, ischemic stroke or transient ischemic attack, and peripheral arterial disease, (ii) post-percutaneous coronary intervention (PCI) with stent or post-percutaneous intervention of peripheral arterial disease, and (iii) prevention of early or acute renal allograft thrombosis in selected high-risk patients in specific centers. However, and unlike other populations, primary prevention is not considered safe choice in KT recipients, due to a poor risk–benefit ratio.

Regarding the choice of antiplatelet, low-dose ASA is the preferred first-line agent in most ESKD patients and KT candidates and recipients, given its well-known pharmacology. Clopidogrel is generally the $P2Y_{12}$ inhibitor of choice when dual antiplatelet therapy (DAPT) is needed (e.g. after acute coronary syndrome) or when ASA is contraindicated. Based on pharmacological profiles of $P2Y_{12}$ inhibitors, ticagrelor requires caution with calcineurin inhibitors (CNIs), particularly with cyclosporin, due to interaction risks. In contrast, such interactions are not noted for clopidogrel or prasugrel (Table 1).

Before initiation of drugs, a full blood count, serum creatinine and estimated glomerular filtration rate (eGFR) (particularly in KT recipients), liver function tests, and a detailed bleeding and thrombotic history, including prior gastrointestinal bleeding and intracranial hemorrhage, should be taken. After initiation of therapy, patients should be closely monitored for signs of bleeding (petechiae, easy bruising, epistaxis, gastrointestinal bleeding, hematuria, menorrhagia), as well as monitoring of hemoglobin/hematocrit and platelet count. In parallel, monitoring of CNI trough levels is needed when ticagrelor is used.

Anticoagulant agents

Anticoagulants encompass four main classes of agents: heparins [unfractionated heparin (UFH), low molecular weight heparin (LMWH) and synthetic pentasaccharide (fondoparinux)],

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Table 1: Oral antiplatelets: mechanism of action, pharmacodynamics and special considerations for KT recipients.

	Aspirin	Clopidogrel	Ticagrelor	Prasugrel	Ticlopidine	Dipyridamole
Mechanism of action	COX-1 inhibitor (direct, irreversible)	ADP $P2Y_{12}$ inhibitor (prodrug, irreversible)	ADP $P2Y_{12}$ inhibitor (direct acting, reversible)	ADP $P2Y_{12}$ inhibitor (prodrug, irreversible)	ADP $P2Y_{12}$ inhibitor (prodrug, irreversible)	Phosphodiesterase inhibitor (↑ cAMP, vasodilatory effect)
Half-life	15–20 min (but platelet inhibition lasts 7–10 days)	6 h (active metabolite effect ~24 h)	7–9 h	7–8 h	12–30 h	10–12 h
Time from last dose to offset	5–7 days (irreversible COX-1 inhibition)	5–7 days (new platelet formation needed)	3–5 days (reversible inhibition)	7–10 days (irreversible inhibition)	7–10 days (irreversible inhibition)	~24 h (shorter effect)
Bioavailability	40%–50% (first-pass metabolism reduces absorption)	>50%	36%	>80%	80%	70%
Recommended dose and frequency	75–100 mg (OD)	300 mg loading dose, then 75 mg OD	90 mg (BID)	10 mg (OD)	250 mg BID	Immediate release: 50–100 mg TID/QID Extended release: 200 mg BID (with aspirin)
Renal excretion	6%–36%	50%	Minimal renal excretion	70%	40%	1%–10%
Interactions	↑ Bleeding risk with NSAIDs, anticoagulants ↓ Effect with ibuprofen (competitive COX-1 inhibition)	↓ Effect with CYP2C19 inhibitors (e.g. PPIs)[a]	↓ Effect with CYP3A4 inducers[b]/↑ effect with CYP3A4 inhibitors		↓ Effect with CYP2C19 inhibitors[d]	↑ Hypotensive effect of antihypertensives Interferes with adenosine clearance
Dialysable/reversal agent	Partially yes/no	No/no	No/no	Unknown/no	No/no	No/aminophylline (to reverse vasodilation in stress testing)
Precautions for transplantation recipients	CNIs: no clinically relevant interaction Mycophenolate: ↑ exposure (due to renal tubular competition)[e]	CNIs: no clinically relevant interaction Mycophenolate: no clinically relevant interaction[e]	CNIs: cyclosporine ↑ ticagrelor exposure; tacrolimus—no relevant effect Mycophenolate: no clinically relevant interaction[b]	CNIs: no clinically relevant interaction Mycophenolate: no clinically relevant interaction[c]	CNIs: no clinically relevant interaction Mycophenolate: ↑ exposure (due to renal tubular competition)[e] Not recommended due to hematological toxicity (neutropenia, thrombocytopenia, TMA)	No

BID, bis in die (twice daily); cAMP, cyclic adenosine monophosphate; COX-1, Cyclooxygenase-1; CYP2C19, Cytochrome P450 2C19; CYP3A4, Cytochrome P450 3A4; NSAIDs, nonsteroidal anti-inflammatory drugs; OD, once daily (omni die); PPIs, proton pump inhibitors; QID, quater in die (four times daily); TID, ter in die (three times daily); TMA, thrombotic microangiopathy. References:

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vitamin K antagonists (e.g. warfarin, acenocoumarol and phenprocoumon), hirudin derivatives (argatroban, bivaluridin and lepirudin) and direct-acting oral anticoagulants. UFH potentiates the activity of antithrombin, which in turn inhibits several coagulation factors, notably factor Xa and factor IIa (thrombin). UFH is generally administered as a continuous intravenous infusion. It has a short half-life of approximately 30 min and can be administered safely to patients with CKD, although those with ESKD necessitate a 33% dose reduction [36]. Its effects can be rapidly and completely reversed by protamine sulfate. Clearance predominantly occurs via endothelial cells and the reticuloendothelial system, with a minor contribution from renal excretion [37]. Adverse effects of UFH include bleeding and thrombocytopenia, with the latter occasionally presenting as HIT, a condition associated with paradoxical thrombosis. UFH's unpredictable pharmacokinetics necessitates monitoring via activated partial thromboplastin time. LMWHs, such as enoxaparin, bemiparin, dalteparin, tinzaparin and nadroparin, also inhibit factor Xa, although they exhibit less efficacy against thrombin compared with UFH. LMWHs have a longer half-life of approximately 4–6 h, better pharmacokinetic properties, and require less frequent monitoring than UFH. Nonetheless, renal elimination necessitates dose adjustment, particularly in advanced CKD patients, owing to an increased risk of bleeding. Routine anti-Xa monitoring is not required in most patients receiving standard weight-based dosing. However, monitoring is recommended in special populations or when dosing deviates from standard protocols, including patients with impaired kidney function, obesity, low body weight, pregnancy and the elderly [36]. Protamine is the widely used and accepted antidote for LMWH, though it does not completely reverse anti-Xa activity.

Vitamin K antagonists (VKAs) such as warfarin, acenocoumarol and phenprocoumon, inhibit the vitamin K epoxide reductase complex 1 and impair the synthesis of coagulation factors II, VII, IX and X, and anticoagulant proteins C and S. They also affect bone metabolism via osteocalcin and matrix Gla protein. The anticoagulant effect of VKAs begins approximately 72–96 h after administration, as the vitamin K–dependent coagulation factors become depleted, and the therapeutic window is relatively narrow. Warfarin's half-life ranges from 24 to 58 h (shorter for acenocoumarol, longer for phenprocoumon). VKAs interact with numerous foods and medications. The quality of anticoagulation is typically assessed through the time in therapeutic range (TTR). However, maintaining an adequate TTR is particularly challenging in dialysis and KT recipients due to fluctuating volume status, changes in nutritional intake, frequent antibiotic exposure and interactions with immunosuppressive agents. Studies have consistently shown that ESKD patients have markedly lower TTR values compared with the general population, leading to higher rates of both subtherapeutic (increasing thrombosis risk) and supratherapeutic (increasing bleeding risk) anticoagulation [38, 39]. Additionally, VKAs have been implicated in promoting vascular calcification progression by inhibiting the vitamin K–dependent matrix Gla protein, which normally suppresses vascular calcification [40], and have also been shown to decrease bone mineral density, leading to an increased risk of fractures [41]. Despite these concerns, many KT recipients with cardiovascular comorbidities may remain on a VKA, although it should be discontinued or bridged with heparin before transplantation surgery [18]. The reversal of VKAs is achieved using vitamin K and, in cases of major bleeding or urgent procedures, prothrombin complex concentrate or fresh frozen plasma may be administered for rapid correction.

Lastly, direct oral anticoagulants (DOACs) represent a more recent advancement in anticoagulation therapy, targeting either direct thrombin or factor Xa in the coagulation cascade. These agents include dabigatran (a direct thrombin inhibitor) and factor Xa inhibitors such as rivaroxaban, apixaban and edoxaban. While DOACs are generally considered safe for patients with CKD stages 1–3, their efficacy and safety in CKD stages 4 and 5 remain uncertain due to limited evidence [42]. Recently, only the use of apixaban was approved by the Food and Drug Administration in patients with GFR <15 mL/min and those on dialysis. Dabigatran is predominantly renally cleared (approximately 85%), rendering it contraindicated in CKD patients with GFR <60 mL/min. Factor Xa inhibitors, characterized by a half-life ranging from 5–12 h, undergo partial renal excretion, albeit to a lesser extent than dabigatran. However, severe CKD may prolong their half-life, necessitating cautious dosing adjustments [36]. There are agents that selectively reverse their actions, such as andexanet-alfa and idarucizumab which antagonize factor Xa inhibitors and dabigatran, respectively [43]. However, their bioavailability seems to be influenced by kidney impairment [44] and caution may be required in ESKD despite preliminary clinical data in CKD patients supporting their efficacy and safety [45].

During the last few years, novel FXI and FXIa inhibitors have emerged, including among others the monoclonal antibodies osocimab and abelacimab, which are not cleared by the kidneys and therefore offer an additional benefit in patients with ESKD [46]. In a recent meta-analysis these drugs did not display any significant increase in bleeding risk compared with placebo among patients with ESKD on HD [47]. However, their use in these patients remains quite limited, while there are currently no data in KT recipients.

Table 2 summarizes the oral anticoagulant agents, mechanisms of action, pharmacodynamics, pharmacokinetics and considerations for KT recipients.

SCORES FOR ASSESSING BLEEDING AND THROMBOTIC RISK

No bleeding or thrombosis risk score has been formally validated in patients with ESKD or in KT recipients/candidates. However, several widely used clinical tools, validated primarily in the general population and partly in advanced CKD, can support decision-making when considering the initiation or continuation of antiplatelet or anticoagulant drugs in this population.

Among bleeding risk scores, the HAS-BLED, originally developed for atrial fibrillation patients receiving anticoagulation, remains one of the most referenced tools [48]. Many of its components [e.g. hypertension, abnormal renal function, labile International Normalized Ratio (INR), age and concomitant antiplatelet use] are highly prevalent in ESKD patients; this significantly limits its discriminatory value. Nevertheless, it offers a structured framework for identifying modifiable bleeding risk factors (e.g. uncontrolled blood pressure, unnecessary DAPT), which can be clinically useful during pre-transplant evaluation or in the peri-operative period [49]. Other bleeding risk scores, ATRIA, HEMORR2HAGES and ORBIT, may not be useful in dialysis patients [50].

Similarly, thromboembolic risk assessment often relies on the CHA_2DS_2-VASc score, widely used for stratifying stroke risk in patients with atrial fibrillation [51]. Its predictive accuracy is reduced in ESKD, because uremia-related alterations in coagulation and vascular biology are not captured by the traditional scoring components. Nonetheless, CHA_2DS_2-VASc remains the

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Table 2: Oral anticoagulants: mechanism of action, pharmacodynamics and special considerations for KT recipients.

	Vitamin K antagonists	FXa inhibitors			Direct thrombin inhibitors
	Warfarin	Rivaroxaban	Apixaban	Edoxaban	Dabigatran
Mechanism of action	Inhibits hepatic synthesis of vitamin K–dependent coagulation factors	Direct inhibition of FXa	Direct inhibition of FXa	Direct inhibition of FXa	Direct inhibition of clot-bound and free thrombin (FIIa)
Half-life (h)	20–60	5–9 (9–13 in elderly)	8–13	9–11	14–17
Time from last dose to offset	3–5 day (INR dependent)	24–48 h (1–2 days)	48 h (2 days)	24–36 h (1–1.5 days)	48–72 h (prolonged in renal impairment)
Bioavailability (%)	100	80	66	50	6.5
Recommended dose and frequency	Adjusted based on INR; OD	20 mg OD	5 mg BID	30 or 60 mg OD	150 mg BID
Renal excretion (%)	1 (unchanged in the urine)	66	27	45	80
Interactions	↑ Effect with CYP2C9/1A2/3A4 inhibitors ↓ Effect with high dietary vitamin K[a]	↑ Effect with potent CYP3A4 inhibitors and P-glycoprotein inhibitors[b]	↑ Effect with potent CYP3A4 inhibitors[b]	↑ Effect with P-glycoprotein inhibitors[b]	↑ Effect with P-glycoprotein inhibitors[b]
Dialysable/drug reversal	No/vitamin K, FFP, PCC	No (only edoxaban has some dialyzability but not highly effective)/andexanet alfa, PCC, 4F-PCC			Partially dialyzable/idarucizumab, PCC, 4F-PCC
Precautions for transplantation recipients	CNIs: tacrolimus has no clinically relevant interaction Cyclosporine may modestly increase warfarin effect[c] Mycophenolate: no clinically relevant interaction	CNIs: cyclosporine markedly increases DOAC exposure; avoid with dabigatran Tacrolimus causes only modest, clinically limited increases Mycophenolate: no clinically relevant interaction Check GFR for DOAC dose modification Prolonged time after cessation is required for patients with low GFR before invasive procedures[d]			

4F-PCC, four-factor prothrombin complex concentrate; BID, bis in die (twice daily); CYP1A2, Cytochrome P450 1A2; CYP2C9, Cytochrome P450 2C9; CYP3A4, Cytochrome P450 3A4; FFP, fresh frozen plasma; FXa, Factor Xa; OD, once daily (omni die); PCC, prothrombin complex concentrate. References:
[a]Montomoli M, Candía BG, Barrios AA. *et al*. Anticoagulation in chronic kidney disease. *Drugs* 2024;84:1199–218 (in English); doi: 10.1007/s40265-024-02077-6.
[b]Ferri N, Colombo E, Tenconi M. *et al*. Drug-drug interactions of direct oral anticoagulants (DOACs): from pharmacological to clinical practice. *Pharmaceutics* 2022;14 (in English); doi: 10.3390/pharmaceutics14061120.
[c]Lam E, Bashir B, Chaballa M. *et al*. Drug interactions between direct-acting oral anticoagulants and calcineurin inhibitors during solid organ transplantation: considerations for therapy. *Expert Rev Clin Pharmacol* 2019;12:781–90 (in English); doi: 10.1080/17512433.2019.1637733.
[d]Lange NW, Muir J, Salerno DM. Direct oral anticoagulants in patients with ESRD and kidney transplantation. *Kidney Int Rep* 2025;10:40–53 (in English); doi: 10.1016/j.ekir.2024.10.016.

most used tool to guide anticoagulation decisions in KT recipients with atrial fibrillation, as it helps quantify baseline thrombotic risk, even if absolute risk estimates may differ from the general population [52].

Clinicians interpreting these scores in ESKD and transplant populations must recognize their limited calibration and the need for individualizing clinical decisions.

METHODOLOGY—SYSTEMATIC SEARCH

This clinical practice document was developed by the Developing Education Science and Care for Renal Transplantation in European States (DEScARTES) scientific working group of the European Renal Association (ERA) in collaboration with experts from European Kidney Transplant Association (EKITA) from the European Society for Organ Transplantation (ESOT) and other content experts (a cardiologist and a hematologist). The group was supplemented with a methodologist to supervise the project and provide methodological expertise. The group identified and constructed four clinical questions. (1) What is the peri-operative bleeding risk in KT candidates who are on antiplatelet or anticoagulant drugs? (2) Should KT recipients receive prophylaxis for venous thrombosis after surgery? (3) Can we use DOACs in KT recipients in terms of safety (dosing, drug–drug interactions, adverse events)? (4) Is there a need for interruption of antiplatelet therapy preceding graft biopsy? To provide evidence-based answers to these questions, a systematic review was performed according to Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) and ERBP principles. A detailed description of methodology is provided in the Supplementary Material.

QUESTION 1: WHAT IS THE PERI-OPERATIVE BLEEDING RISK IN KT CANDIDATES WHO ARE ON ANTIPLATELET OR ANTICOAGULANT DRUGS?

Authors: Rachel Hellemans, Ilaria Gandolfini, Rachel Thomas, Artemios Karagiannidis, Arzu Velioglu

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Background and rationale

In order to optimize the safest care for KT candidates on antithrombotic therapies, clinicians should have a comprehensive knowledge of the possible consequences, such as risk of perioperative bleeding. There is little antithrombotic management data for KT surgery, mostly extrapolated from other specialties. In general, antiplatelet or anticoagulant drugs should be stopped to reduce the risk of bleeding complications, however, in deceased donor transplantation, an inherently unplanned surgery, there is often insufficient time for drug effects to subside, and delaying the procedure is undesirable due to concerns about cold ischemia time and graft viability. At the same time, interrupting antithrombotic therapy may increase the risk of thromboembolic events, especially in patients with coronary artery disease, a history of stroke and previous VTE. Therefore, we aimed at reviewing the impact of continuation or discontinuation of pretransplant antiplatelet or anticoagulant drugs on peri-transplant bleeding and thromboembolic complications.

Summary of evidence

(Supplement | Study selection flow diagram for Question 1, see Supplementary data, Fig. S1)

(Supplement | Summary evidence table for Question 1, see Supplementary data, Table S10)

Given the highly diverse outcomes, definitions, and variable rates of complications in this systematic review, we had to review evidence categorized by the type of antithrombotic therapy. In any case, as the studies were heterogeneous in design, population and reported outcomes, overall risk of bias was high and certainty of evidence was low.

Antiplatelet therapies

Our search identified eight original studies (five full papers, three abstracts) and one systematic review and meta-analysis that investigated the impact of antiplatelet drugs on post-operative outcomes. These included a small randomized controlled trial and seven single-center retrospective cohort studies.

Antiplatelet monotherapy versus no antiplatelet therapy

Aspirin

Robertson *et al.* [53] found a significant risk reduction in rates of early renal vein graft thrombosis with perioperative ASA (pretransplant to 1-month posttransplant) as compared with a historical cohort without ASA, but noted early major bleeding events, although these were not compared with controls and confounding factors (e.g. patient selection, improved care), limiting the conclusions. Rodriguez-Pastore *et al.* [54] conducted a trial of 30 recipients on ASA at time of transplantation. The patients were randomized to receive a perioperative platelet transfusion or not and results were compared with a control group of 70 patients not on ASA. They found similar hemoglobin drop across all three groups, but higher rates of ultrasound-detected perirenal bleeding in ASA patients. The small sample size and limited data preclude firm conclusions.

Clopidogrel

Kapeles *et al.* [55] found significantly higher perioperative transfusion rates in pretransplant clopidogrel users, though details were limited making it difficult to assess the study's reliability or relevance.

DAPT versus no antiplatelet therapy

Bansal *et al.* [56] reported a higher drain output in patients on DAPT at transplantation compared with patients who stopped DAPT 5 days pre-transplant, with other perioperative outcomes (blood loss, transfusion, hemoglobin drop and cardiovascular events) being similar between groups.

DAPT versus single antiplatelet therapy

A systematic review and meta-analysis by Lee *et al.* [57] including five retrospective studies [26, 58–61] showed DAPT increased post-operative hemorrhage risk almost by 60% in patients on DAPT compared with those on monotherapy; cardiovascular event rates were similar, though not quantitatively analyzed. The studies were small, heterogeneous and of low quality, with inconsistent definitions of hemorrhagic events, limiting the ability to assess the clinical relevance.

Mixed or unspecified antiplatelet therapies versus no therapy

Hernández *et al.* [62] reported higher hemorrhagic complication rates among patients on antiplatelet therapy at transplantation compared with controls, with multivariable analysis confirming this 1.8-fold higher odds of the outcome, but the lack of specification regarding the type of antiplatelet therapy weakens the study's conclusions. Benahmed *et al.* [63] found similar rates of transfusion, reoperation or cardiovascular events between antiplatelet users (clopidogrel/ticlopidine ± ASA) and controls on no therapy; however, the sample was small and treatment variability limit the study's conclusions.

Anticoagulant therapies

Our search identified five original studies (three full papers, two abstracts). These included a small randomized controlled trial, two retrospective cohort studies and two case series.

Heparin

Kikić *et al.* [64] randomized adult preoperative patients to different preoperative dialysis anticoagulation approaches (i.e. either heparin HD or no dialysis in case of pre-operative potassium ≤5 mEq/L, and to either citrate HD or heparin HD in case of preoperative potassium >5 mEq/L). They reported bleeding events in all groups, with no significant differences between heparin and citrate or no-heparin strategies. Despite moderate evidence quality, intradialytic heparin data does not generalize to anticoagulant use.

VKA—warfarin

Connaughton *et al.* [65] examined warfarin-treated recipients compared with sex-matched controls, reversing warfarin preoperatively per guidelines. No differences emerged in hemoglobin, graft or patient survival, but warfarin patients required more perioperative blood products (transfusion rates, fresh frozen plasma use). Interpretation is limited by the small size and poor matching. Similarly, Wong *et al.* [66] reported that warfarin users compared with matched controls tended to be older, with higher prevalence of ischemic heart disease, longer hospitalization and higher transfusion rates. Post-transplant anticoagulation increased reoperation risk. However, the study's limited reporting substantially weakens the reliability of these findings.

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DOACs—apixaban

Two case series reported the early post-transplant outcomes of deceased-donor recipients on pre-transplant apixaban anticoagulation therapy. Littlejohn *et al.* [67] described a small cohort with variable apixaban dosing (5 mg, $n = 7$; 2.5 mg, $n = 5$; high-dose with aspirin, $n = 3$) and noted several early bleeding events. Lange *et al.* [68] similarly evaluated a small number of KT recipients on apixaban with variable doses and no reversal agents. Blood loss ranged from 100 to 1600 mL and notable bleeding complications were observed (major vascular injury, posttransplant transfusions) with high bleeding rates. Taken together, studies suggest possible increased rates of bleeding with apixaban at the time of transplantation, yet variations in dosage, small sample sizes and the absence of control groups make it difficult to draw definitive conclusions.

Miscellaneous studies (mixed therapies of VKAs, unfractionated heparin, antiplatelet agents)

Additionally, we identified 11 studies (9 full papers, 2 abstracts). These included eight retrospective cohort studies, one case–control study and two abstract papers that compared a variety of antithrombotic therapies in patients undergoing kidney transplantation.

van den Berg *et al.* [25] investigated bleeding events within the first week post-transplant across multiple treatment groups (control, intraoperative heparin, continued VKA, continued antiplatelet therapy, combined antiplatelet and heparin) and found that continued VKA use was independently associated with an increased risk of post-operative bleeding, reporting more than 6-fold higher odds compared with controls. Similarly, Marzouk *et al.* [69] observed higher transfusion rates among patients receiving pre-operatively warfarin or antiplatelet agents compared with those not on such therapies, a finding that was also confirmed by Musetti *et al.* [59] reporting that patients receiving pre-transplant VKA had higher rates of post-operative bleeding compared with those receiving antiplatelet therapies and controls. In a more recent study, Hau *et al.* [49] also found that preoperative use of VKA significantly increased the risk of post-operative bleeding by almost six times, but antiplatelet agent use did not.

Weng *et al.* [61] analyzed KT recipients with coronary artery disease and found that patients using DAPT had higher transfusion rates at transplant, although rates of hematoma drainage were similar to recipients on other pre-transplant antithrombotic regimens.

In contrast, numerous studies did not demonstrate a clear increase in bleeding complications with antiplatelet or anticoagulant use. Eng *et al.* [70] reported no significant increase in reoperation, transfusions or hemoglobin drop in patients receiving preoperative antiplatelets or anticoagulants. Another retrospective study comparing recipients on antiplatelets or anticoagulants with controls found that ASA did not significantly alter bleeding risk, whereas anticoagulants were associated with a higher risk of bleeding events [71]. In parallel, Diamond *et al.* [72], examining a large cohort of KT recipients divided into three groups (anticoagulant users, antiplatelet users, controls), found no significant between-group differences in major bleeding or thromboembolic events. Alonso-Escalante *et al.* [73] found comparable rates graft re-exploration rates and perioperative transfusions between pre-transplant warfarin users, antiplatelet users and untreated controls.

A case–control study including recipients with and without post-operative hemorrhage found no clear association between pretransplant antiplatelet or anticoagulant use and post-operative hemorrhage risk [26]. Finally, Buggs *et al.* [74] found that recipients with post-operative bleeding requiring reoperation did not differ in their preoperative anticoagulation status from those who did not require re-operation.

Taken together, these studies consistently suggest that perioperative use of VKAs is associated with a higher risk of bleeding complications, while DAPT may also increase transfusion needs at the time of KT.

Translation of the evidence into clinical practice

The available evidence, despite its very low quality, indicates that clopidogrel or DAPT probably increases the risk of perioperative bleeding. However, the extent and clinical significance of this risk remain poorly defined. Data regarding the perioperative risks associated with aspirin monotherapy are also limited, but there is a trend indicating increased risk of bleeding or need for transfusion.

Studies on VKAs suggest an increased bleeding risk, particularly when anticoagulation is continued post-transplant. Perioperative INR values, suggestive of TTR, were not consistently reported in the studies, and therefore we cannot make any recommendation about safe threshold values or reversal agents. Additionally, data on DOACs is too limited to draw definitive conclusions and therefore cannot be supported in deceased kidney transplantation.

What do existing guidelines say?

The KDIGO 2020 guidelines [3] state that KT candidates should not be excluded from transplantation due to anticoagulation or antiplatelet medication. Single antiplatelet agents (e.g. aspirin) can be continued while on the deceased donor waiting list. Transplant should be delayed for patients on DAPT (e.g. aspirin + clopidogrel) if the risks outweigh the benefits. KDIGO also recommends stopping antiplatelet agents (except for aspirin) 5 days before elective live donor transplants. Warfarin should also be stopped 5 days prior for elective transplants, with bridging as needed. DOACs (e.g. apixaban) are discouraged unless there is perioperative expertise and reversal options and therefore, most patients should be switched to alternatives (e.g. warfarin) if needed.

Current European Society of Cardiology (ESC) guidance on perioperative antithrombotic management in non-cardiac surgery provides several principles that may be applicable to KT surgery, classified as a moderate cardiac and high bleeding risk surgery [75]. In patients with a prior PCI, continuation of low-dose aspirin throughout the perioperative period is recommended when the surgical bleeding risk is acceptable (Class I). For candidates without a history of PCI, aspirin discontinuation at least 3 days before transplantation may be considered when bleeding risk clearly exceeds ischemic risk (Class IIb), given the substantial intraoperative and post-operative bleeding burden associated with transplant surgery. Elective procedures should ideally be postponed for at least 6 months after PCI (Class I). Once the mandatory DAPT period has passed, $P2Y_{12}$ inhibitors should be withheld according to their pharmacodynamic profiles (3–5 days for ticagrelor, 5 days for clopidogrel and 7 days for prasugrel) prior to surgery (Class I). For patients on anticoagulants who are at high thrombotic risk, perioperative bridging may be considered, otherwise bridging is not recommended (Class IIb).

According to the CHEST perioperative guidelines published in 2022 [76], KT is acknowledged as a high-bleed-risk surgery (quoted as >2% 30-day risk of major bleeding). This guideline does not also recommend heparin bridging for patients on oral

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Table 3: Summary of recommendations for pre-transplant antithrombotic use and management before surgery.[a,b]

Antithrombotic therapy	Living donor kidney transplantation	Deceased donor kidney transplantation
Antiplatelets (monotherapy)	Stop non-aspirin antiplatelets 5–7 days before KT. Aspirin may be continued unless bleeding risk is excessive	Can be continued while awaiting KT. Stop during the perioperative period. Aspirin may be continued unless bleeding risk is excessive
DAPT	Delay KT for the mandated period of treatment with DAPT after PCI. Once completed, continue aspirin, stop the $P2Y_{12}$ inhibitor (ticagrelor 3–5 days; clopidogrel 5 days; prasugrel 7 days)	Proceed if KT benefits outweigh bleeding or thrombosis risks. Otherwise delay until the mandatory DAPT period is completed. Convert monotherapy during waitlisting whenever clinically feasible
VKAs	Stop 5 days before KT. Consider heparin/LMWH bridging only in patients with high thrombotic risk	Administer reversal agents (vitamin K, FFP, PCC) with awareness of increased bleeding risk
DOACs	Stop ≥48–72 h before KT	Proceed only if drug level measurement and reversal agents are available. Per center experience and hematologist consultation are needed. Otherwise not preferred during waitlisting

FFP, fresh frozen plasma; PCC, prothrombin complex concentrate. Note: **bold** for conditional cases, *italics* for high-risk situations.
References:
[a]Chadban SJ, Ahn C, Axelrod DA. *et al*. KDIGO Clinical Practice Guideline on the Evaluation and Management of Candidates for Kidney Transplantation. *Transplantation* 2020;104(4S1 Suppl 1):S11–103; doi: 10.1097/TP.0000000000003136.
[b]Halvorsen S, Mehilli J, Cassese S. *et al*.; ESC Scientific Document Group. 2022 ESC Guidelines on cardiovascular assessment and management of patients undergoing non-cardiac surgery. *Eur Heart J* 2022;43:3826–924; doi: 10.1093/eurheartj/ehac270.

anticoagulant treatment before surgery depending on the indication. However, there are no further data offered to clarify additional risk; this guideline only covers elective surgery, and it is unknown whether these recommendations can be safely extrapolated to KT surgery (both elective and emergency).

These recommendations are summarized in the Table 3.

Clinical practice points

- The decision to continue, interrupt or modify antithrombotic therapy prior to transplantation should be individualized based on the expected waiting time for transplantation, availability of reversal agents and the center's experience.
- Clinicians should inform patients about the potential risk of bleeding with any antithrombotic therapy used in deceased donor KT, which is significant for clopidogrel, DAPT and anticoagulants.
- DOAC use in waitlisted patients is not recommended, given the limited evidence regarding their safety in transplant surgery.

QUESTION 2: SHOULD KT RECIPIENTS RECEIVE PROPHYLAXIS FOR VENOUS THROMBOSIS AFTER SURGERY?

Authors: Erol Demir, Luuk Hilbrands, Artemios Karagiannidis

Background and rationale

Venous thrombosis is a significant risk factor for early allograft loss and contributes to perioperative morbidity in KT recipients. To prevent such adverse outcomes in the post-operative period, clinicians use thromboprophylaxis regimens, but the exact protocols, timing and eligibility criteria differ between centers, because no universal guidelines are available, and this decision is mostly based on individual center experience. Therefore, we aimed to evaluate the efficacy of thromboprophylaxis post KT to prevent venous thrombosis and pulmonary embolism and summarize the exact details of such regimens. Specifically, we reviewed evidence regarding the optimal timing, appropriate dosing and overall effectiveness of thromboprophylaxis, as well as its potential risks.

Summary of evidence

(Supplement | Study selection flow diagram for Question 2, see Supplementary data, Fig. S2)

(Supplement | Summary evidence table for Question 2, see Supplementary data, Table S11)

Given the highly diverse outcomes, definitions and variable rates of complications, we opted again to review evidence categorized by the type of anticoagulant medication. Our search identified 10 original studies (2 randomized prospective clinical trials, 1 prospective cohort study and 7 retrospective cohort studies). The studies were heterogeneous in design, population and outcomes, with overall high risk of bias and low certainty of evidence.

Antithrombotic prophylaxis with unfractionated heparin or low-molecular weight heparin

In a single-center prospective study [77], patients were randomized to receive either no heparin or subcutaneous (SC) unfractionated heparin for 7 days or until the patient achieved fully mobility. Within 18 days post-operatively, thromboembolic events (graft vein thromboses, arteriovenous fistula thromboses, deep vein thrombosis and pulmonary embolism) were reported only in the no-heparin group. Although the study was small and not powered for statistical significance, the findings suggest a protective effect of post-operative heparin prophylaxis without significant increases in bleeding or wound complications.

In a prospective cohort study [78], patients without hypercoagulable states (normal coagulation screening tests, absence of a clinical history of hypercoagulability) received dalteparin immediately after surgery and then daily throughout their hospitalization. Only one case of venous thrombosis was reported, and no bleeding complications were reported. The lack of a control group limits the assessment of the true efficacy of this prophylactic approach.

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A third prospective study randomized living donor KT recipients to receive either unfractionated heparin for 1 week, tinzaparin for 1 week or no prophylaxis [79]. No thromboembolic events were reported in any group, limiting the assessment of the potential benefit of prophylaxis. However, decreases in post-operative hemoglobin levels were higher in the anticoagulated groups (with higher decrease in the unfractionated heparin group compared with the LMWH group), highlighting potential associations between anticoagulation and mild hemoglobin reduction.

In a retrospective cohort study [80], patients who received peri- or post-operative enoxaparin were compared with those who did not; prophylaxis was administered in the first group for a mean duration of 10.4 days. One venous graft thrombosis was reported in the no-prophylaxis group, but there were no detailed data on the incidence of deep venous thrombosis and pulmonary embolism. In a second retrospective study [81] comparing various prophylactic heparin regimens (heparin 5000 IU SC twice daily, heparin 5000–7500 IU SC thrice daily, dalteparin 5000 IU SC daily) with no prophylaxis, thrombotic events during the first post-operative week were uncommon in both groups, with slightly fewer events among those receiving prophylaxis. Importantly, bleeding risks were comparable between groups in both studies, a finding that further supports the safety of both unfractionated and LMWH prophylaxis in KT recipients.

In a large retrospective cohort study [82], deep vein thrombosis (excluding graft thrombosis) occurred in almost 5% of KT recipients within the first 3 months after transplantation. Antithrombotic prophylaxis, defined as at least one dose of SC unfractionated heparin 5000 IU administered between 24 h before and 48 h after transplantation, did not reduce the risk of deep vein thrombosis. However, the retrospective design and the lack of precise information on prophylaxis duration preclude definitive conclusions.

Finally, van den Berg *et al.* [25] showed that post-operative continuous administration of unfractionated heparin (12 000 IU/day for 5 days) resulted in a higher bleeding risk than prophylactic SC nadroparin (2850 IU/daily), underscoring the importance of avoiding unnecessarily aggressive anticoagulation regimens.

Antithrombotic prophylaxis with aspirin

A retrospective study demonstrated a lower incidence of renal vein thrombosis after prescribing low-dose aspirin (75 mg/day), initiated immediately before and continued for 1 month after transplantation. However, confounding factors (e.g. patient selection, improved care over time between cohorts) limit the strength of this observation [53]. Importantly, patients in this study did not receive heparin prophylaxis. Another retrospective study found no cases of primary allograft thrombosis among patients receiving higher-dose aspirin (150 mg/day) for the first three post-operative months combined with a short course of peri-operative heparin (5000 IU pre-transplantation and 5000 IU twice daily SC for 5 days post-transplantation) [83]. Cases of graft thrombosis were observed only in the untreated control group, with an incidence higher than what is currently reported in patients receiving LMWH prophylaxis.

In a retrospective single-center study [84], the duration of aspirin treatment (100 mg/day for 5 days or >6 weeks post-operatively) did not appear to affect VTE incidence in the first 6 weeks following KT. In this study, patients could receive concurrent heparin prophylaxis, but the exact proportion of patients who received heparin was not reported.

Translation of the evidence into clinical practice

While LMWH prophylaxis for venous thrombosis is routinely used in many KT centers, solid support for this practice is lacking. In patients without specific risk factors for venous thrombosis, the absolute incidence of thromboembolic events after kidney transplantation is low despite an increased relative risk.

Of the three small prospective studies, one did not include a control group and in another one there were no cases of deep vein thrombosis in the control group, which precludes conclusions about efficacy. The retrospective studies included more patients but were low-quality evidence. Nevertheless, there was not a substantially increased risk of bleeding to be associated with antithrombotic prophylaxis.

There is no strong evidence on a potential additional beneficial effect of aspirin in patients receiving LMWH prophylaxis in whom the rate of venous allograft thrombosis is already low.

Taken together, based on the known increased incidence of VTE after KT, the proven efficacy in other types of pelvic and vascular surgery, and the favorable safety profile of prophylactic doses of LMWH in KT recipients, current practice appears to be justified.

What do existing guidelines say?

The European Association of Urology Guidelines on Renal Transplantation (2024) [85] provides the following recommendation, graded as weak: "Do not routinely give post-operative prophylactic unfractionated or low-molecular weight heparin to low-risk living donor transplant recipients." The recommendation is based on a single study involving 75 live donor kidney recipients [79], as discussed earlier.

The following guidelines do not give recommendations on routine prophylaxis for venous thrombosis after surgery:

- Post-operative care in the KT recipient of the UK Kidney Association (2017) [86]
- KDIGO clinical practice guideline for the care of KT recipients (2009) [87]

Clinical practice points

- Based on the efficacy of antithrombotic prophylaxis in other surgical settings, antithrombotic prophylaxis after kidney transplantation can be justified.
- Prophylaxis with unfractionated heparin or LMWH is not associated with an increased risk of bleeding.

QUESTION 3: CAN WE USE DOACs IN KT RECIPIENTS IN TERMS OF SAFETY (DOSING, DRUG–DRUG INTERACTIONS, ADVERSE EVENTS)?

Authors: Ivana Dedinská, Annelies de Weerd, Artemios Karagiannidis, Matej Samoš

Background and rationale

The long-term use of oral anticoagulation in KT recipients with atrial fibrillation/atrial flutter (AF/AFlu) and VTE is crucial to prevent thromboembolic events (stroke or systemic embolism in those with AF/AFlu and repeated pulmonary embolism and/or deep vein thrombosis in those with VTE). However, there are no specific recommendations and guidelines on therapeutic strategies in these patients. Traditional oral anticoagulants like VKAs, such as warfarin, present challenges due to narrow therapeu-

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tic indices, the need for frequent monitoring, risk of calciphylaxis and potential interactions with immunosuppressants or some other drugs, such as antibiotics. DOACs, including apixaban, rivaroxaban, dabigatran and edoxaban, have an alternative mechanism of action and have become first-line treatment for AF and the prevention of thromboembolism in the general population (except if mitral stenosis or mechanical heart valve) [88]. DOACs offer a potentially safer alternative to VKAs due to their predictable pharmacokinetics and reduced need for monitoring. However, although there is no clear contraindication for DOACs therapy in KT recipients, their use requires careful consideration due to several factors:

(i) Kidney function: KT recipients often have impaired graft function, which can affect DOACs clearance and increase the risk of bleeding
(ii) Drug–drug interactions: DOACs can interact with immunosuppressants, particularly CNIs, tacrolimus and cyclosporine, potentially altering their pharmacokinetic and pharmacodynamic profiles
(iii) Bleeding risk: it associated with DOACs remains a critical concern due to altered kidney function, potential drug–drug interactions with immunosuppressants, particularly during invasive procedures
(iv) Limited data: compared with their use in other populations, evidence on DOAC use in KT recipients is limited, especially regarding long-term safety and optimal dosing strategies

Therefore, we aimed at evaluating the efficacy of DOACs on KT recipients and reviewing evidence regarding the appropriate dosing, exact drug–drug interactions, overall effectiveness and adverse effects.

Summary of evidence

(Supplement | Study selection flow diagram for Question 3, see Supplementary data, Fig. S3)

(Supplement | Summary evidence table for Question 3, see Supplementary data, Table S12)

The evidence regarding DOAC use in KT recipients is heterogeneous and generally of low certainty, making definitive recommendations challenging. Our search identified 15 studies (10 retrospective studies, 1 open-label drug–drug interaction study and 4 prospective studies).

DOAC choice and dosage

Apixaban studies suggest lower major bleeding rates compared with warfarin, although these findings are not consistently replicated across all studies and are often based on small sample sizes and/or retrospective designs [89–94]. The studies show varying results with rivaroxaban, with some showing no increased bleeding risk compared with warfarin [95, 96], while other reports a higher incidence of minor bleeding [97]. Data on dabigatran and edoxaban are limited, preventing robust conclusions about their safety and efficacy in KT recipients.

Bleeding and thrombotic events

There are no randomized controlled trials comparing DOAC versus VKA in KT recipients. We identified five studies comparing DOACs and VKAs in terms of bleeding and thromboembolic events [89, 90, 95, 96, 98] (Table 4). Four were retrospective studies, only one of which was multicenter prospective follow-up study [90]. With regards to safety, Leon *et al*. reported significantly less bleeding events with DOACs [98], whereas three studies a non-significant trend to less bleeding [89, 95, 96] and one study had more bleeding events [90].

Regarding efficacy, in two studies there were no thromboembolic events, two studies reported non-significantly less and one study non-significantly more thromboembolic events [89, 90, 95, 96, 98]. A limitation of all studies was the lack of information concerning the selection of anticoagulation, which was at the discretion of the treating physician. The duration of the study period additionally impacted outcomes, as most studies had shorter follow-up durations for DOACs. Four studies explicitly excluded patients with eGFR <30 mL/min [90, 96, 98, 99], one study did not report kidney function [95] and one study described that 7% of DOAC and 15% of warfarin treated patients had eGFR <30 mL/min [89].

Do *et al*. conducted a single-center retrospective study of 52 adult KT recipients receiving apixaban, reporting two major bleeding events (3.8%) along with two thromboembolic events (3.8%) over 12 months. [100] The lack of a comparison group makes it difficult to interpret the outcomes.

Drug–drug interactions

Several studies highlight potential interactions between DOACs and immunosuppressants (particularly CNIs), leading to altered drug levels [91, 92, 101, 102]. In contrast Camporese *et al*. [99] conducted a prospective study on eight KT recipients receiving tacrolimus (with or without everolimus) and rivaroxaban for VTE or non-valvular AF. Over a 2-week period, no major bleeding or thromboembolic events occurred, and tacrolimus/everolimus through levels remained stable, suggesting no significant drug–drug interactions. A single-center retrospective study on 31 KT recipients receiving DOACs (mostly apixaban) showed that 94% of measured DOAC levels were within the expected therapeutic range [94]. In another , Zaitoun *et al*. [96] found no significant changes in CNI trough-to-dose ratios in patients receiving DOACs.

Overall, drug–drug interactions are often not clinically significant in terms of adverse events, but further research is needed to fully understand the impact and to determine whether therapeutic drug monitoring is routinely beneficial.

Other outcomes

The impact of DOACs on graft function (eGFR) is unclear, with some studies reporting no significant effect, while others show possible improvement with rivaroxaban [91, 92]. Furthermore, mortality rates are not consistently compared between patients on DOACs versus those on warfarin.

Translation of the evidence into clinical practice

Because of the limitations of available evidence, particularly the lack of randomized controlled trials or large-scale propensity score–matched studies, the decision to use DOACs in KT recipients should be individualized, carefully weighing the potential benefits against the risks. However, current findings from existing studies support the use of DOACs based on their effective and safety profile.

It should be noted that there is still a paucity of data in patients with an eGFR <30 mL/min/1.73 m², requiring extra caution in this group. The most clinical experience to date has been obtained with apixaban. The choice of DOAC and dosage should be guided by the patient's clinical characteristics and graft function. Due to the complexity of this patient population who of-

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Table 4: Studies for safety of DOACs comparing with warfarin in KT recipients.

Study (author, year)	Country	DOAC	VKA	Characteristics	Follow-up	Bleeding event	Thromboembolic event	Remark
Bixby *et al.*, 2020[a]	USA	N = 99 (apixaban, *n* = 60; dabigatran, *n* = 6; rivaroxaban, *n* = 33)	Warfarin, N = 98	Median 6.5 years posttransplant	11 vs 13 months	Major bleeding: 9% vs 21%	VTE: 2% vs 3%; stroke: 1% vs 5%	eGFR <30 mL/min: 7% vs 15%
Leon *et al.*, 2020[b]	France	N = 52 (apixaban, *n* = 36; dabigatran, *n* = 1; rivaroxaban, *n* = 15)	N = 50 (warfarin, *n* = 20, fluindione, *n* = 30)	87 vs 65 months posttransplant	14 vs 22 months	Bleeding: 13% vs 42% (P = .002)	PE or DVT: 0% vs 8%	No DOAC if mechanical valve
Zaitoun *et al.*, 2021[c]	Saudi Arabia	N = 8 (apixaban, *n* = 1; dabigatran, *n* = 1; rivaroxaban, *n* = 6)	Warfarin, N = 8 (matched)	Not specified	Unclear	Bleeding: 0% vs 38%	0% vs 0%	Limited sample size
Firth *et al.*, 2023[d]	USA	N = 208 (apixaban, *n* = 189; rivaroxaban, *n* = 19)	Warfarin, N = 320	DOAC more recent (post-2016)	449 vs 1098 days	Major bleeding: 5% vs 8%	VTE: 36% vs 39%; ischemic stroke 5% vs 5%	eGFR not reported
Santoro *et al.*, 2024[e]	Italy	N = 66 (apixaban, *n* = 32; edoxaban, *n* = 17; dabigatran, *n* = 3; rivaroxaban, *n* = 14)	Warfarin, N = 50	Median 145 vs 90 months posttransplant	20 months	Major bleeding: 6% vs 4%	0% vs 0%	AF indication: 71% vs 56%

DVT, deep vein thrombosis; PE, pulmonary embolism.

References:
[a]Bixby AL, Shaikh SA, Naik AS. *et al*. Safety and efficacy of directE: Venous Thromboembolismicoagulants, lar assessment and management of patients undergoing non-cardiac surgery. erwise no*Transpl Int* 2020;33:740–51; doi:10.1111/tri.13599
[b]Leon J, Sabbah L, Aubert O. *et al*. Efficacy and Safety of direct oral anticoagulants in kidney transplantation: a single-center pilot experience. *Transplantation* 2020;104:2625–31; doi: 10.1097/TP.0000000000003168.
[c]Zaitoun MF, Sheikh ME, Faifi ASA. *et al*. The use of non–vitamin K antagonist oral anticoagulants in post-kidney transplantation, single-center experience. *Transplant Proc* 2021;53:2918–22; doi:10.1016/j.transproceed.2021.09.042.
[d]Firth C, Shamoun F, Apolinario M. *et al*. Safety and mortality outcomes for direct oral anticoagulants in renal transplant recipients. *PLoS ONE* 2023;18:e0285412; doi:10.1371/journal.pone.0285412.
[e]Santoro F, Casanova A, Simone S. *et al*. Immunosuppressive therapy and oral anticoagulation in kidney transplant recipients: direct oral anticoagulants versus vitamin-K antagonists. *Eur J Intern Med* 2024;119:71–7; doi:10.1016/j.ejim.2023.08.003.

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ten have altered pharmacokinetics, multiple comorbidities, and concomitant immunosuppressive therapies, management of anticoagulation should involve interdisciplinary collaboration, especially between transplant specialists, cardiologists or vascular specialists and clinical pharmacists to optimize safety and therapeutic efficacy.

What do existing guidelines say?

There is no universal or specific guideline that directly addresses the use of DOACs in KT recipients.

Clinical practice points

- The choice of anticoagulant (DOAC vs VKA) should be individualized; although existing studies suggest that DOACs have a favorable efficacy and safety profile, the evidence in KT recipients remains insufficient to recommend one class over the other.
- Close monitoring of kidney function is recommended for KT recipients on DOACs, given the need for dose modification.
- The use of DOACs in KT recipients with GFR <30 mL/min/1.73 m² should be approached with caution.
- DOACs may interact with CNIs, though most interactions appear clinically insignificant.

QUESTION 4: IS THERE A NEED FOR INTERRUPTION OF ANTIPLATELET THERAPY BEFORE GRAFT BIOPSY?

Authors: Erol Demir, Artemios Karagiannidis, Andreas Kousios

Background and rationale

Bleeding is a significant complication of kidney graft biopsies, with the potential to impact graft and patient outcomes. We aimed at investigating the necessity and safety of interrupting antiplatelet therapy prior to kidney graft biopsy to mitigate bleeding risks. Specifically, we reviewed current evidence on the clinical rationale, patient-specific considerations and potential risks associated with therapy interruption, as well as strategies to balance bleeding and thrombotic complications.

Summary of evidence

(Supplement | Study selection flow diagram for Question 4, see Supplementary data, Fig. S4)

(Supplement | Summary evidence table for Question 4, see Supplementary data, Table S13)

Our search identified seven original studies, all of which were of retrospective nature.

Robertson *et al.* [53] investigated bleeding complications in KT recipients receiving either 75 mg or 150 mg of aspirin daily for at least 1 month to prevent renal vein thrombosis. Post-biopsy bleeding occurred in a small proportion of patients, representing 1.6% of all biopsies, a finding consistent with rates reported by prior studies (0.3%–12.6%). However, the study's interpretability is limited by the lack of clarity about whether aspirin was continued at the time of graft biopsy; this undermines the ability to accurately contextualize the bleeding risk.

Murphy *et al.* [83] evaluated deceased donor KT recipients treated with aspirin 150 mg daily for the first 3 months post-transplant and compared them to a historical control group without aspirin. All patients underwent protocol needle-core biopsies at 1 week and 12 months after transplantation. Rates of macroscopic hematuria following graft biopsy were similar between groups; however, the lack of information on how many biopsies were performed while patients were actively taking aspirin is a strong limitation.

A single-center, 5-year retrospective study of >2500 renal transplant biopsies examined major complications and associated risk factors. [103] Anticoagulants and antiplatelets (warfarin, clopidogrel, and aspirin low- and full-dose), were withheld for 7 days, unless an urgent biopsy was required. Major complications within 14 days post-biopsy were uncommon. Several clinical parameters, i.e. older age, lower platelet count, higher blood urea nitrogen (BUN), deceased donor transplant, prior renal transplant, and use of heparin, clopidogrel or warfarin were recognized as significant risk factors. Cutoff values for increased complication risk included age ≥51 years, creatinine ≥2.23 mg/dL, BUN ≥40 mg/dL, platelet count ≤173 000/μL and hematocrit ≤31%. In contrast, aspirin use was not associated with higher risk for biopsy complications.

In another single-center retrospective study of >6000 renal allograft biopsies, Baffour *et al.* [104] examined the association between aspirin use and post-biopsy bleeding event. All ultrasound-guided biopsies were performed with INR <1.5 and platelet counts $>50 \times 10^9$/L, using ≥18-gauge needles, with three cores obtained per patient. Most of the performed biopsies were protocol biopsies (81.3%). Aspirin doses were low (81 mg/day) or high (325 mg/day). Overall bleeding rates were low; bleeding occurred in 1.7% of patients who took aspirin within 3 days of biopsy, 0.3% in those who took it 4–7 days before, and 0.3% in those who took it 8–10 days prior, compared with 0.7% in non-users. Importantly, there was no association between aspirin use and bleeding risk, except in patients on high-dose aspirin taken within 3 days of biopsy, who presented higher risk of both minor and major bleeding.

Li *et al.* [105] developed a standardized protocol for percutaneous kidney graft biopsies to reduce bleeding complications by optimizing blood pressure, platelet function and anticoagulation management. Implementation of this protocol led to a notable reduction in overall and major hemorrhage rates by almost 50% and 70%, respectively. The approach included real-time ultrasound guidance and discontinuation of aspirin and clopidogrel for 5–7 and 7 days, respectively, before elective biopsies. While effective, the protocol's applicability is limited by its single-center setting and variability of patients' characteristics. Kuiper *et al.* [106] assessed the predictive value of hemostasis tests and clinical factors for bleeding complications in patients undergoing kidney graft biopsy, 26 (17%) of whom were on aspirin. Aspirin was discontinued in all but one patient 5 days before the biopsy. Although overall bleeding rates were low, with only one major complication requiring transfusion, prior use of aspirin (discontinued 5 days before biopsy) was linked to higher bleeding risk. The study concluded that routine hemostasis testing is unreliable for predicting post-biopsy bleeding, with aspirin use being the primary risk factor.

Lastly, Mattiazzi *et al.* [107] compared outcomes of ultrasound-guided kidney graft biopsies performed by either interventional radiologists or transplant nephrologists. Antiplatelet or anticoagulant drugs were used by 40% and 4% of the patients, respectively, and was routinely discontinued at least 7 days before the biopsy. Complication rates were similar between the two groups, with uncontrolled blood pressure and prior antiplatelet use linked to higher risk of post-biopsy complications. These findings underscore the need for individualized management, particularly in optimizing blood pressure control and timing of

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antithrombotic interruption to balance bleeding and thrombotic risks in KT recipients.

Translation of the evidence into clinical practice

Overall, the current literature suggests that major bleeding following allograft biopsy remains rare, even in patients with recent aspirin exposure. However, due to limited evidence and the retrospective nature of existing studies, no clear consensus exists regarding peri-procedural management of antiplatelet drugs in KT recipients. Most transplant centers likely adopt a case-by-case approach, performing individualized risk-benefit analyses, particularly in patients with recent cardiovascular interventions such as coronary stents. Given the recognized high bleeding risk associated with percutaneous biopsy, these decisions must balance thrombotic risk against the potential for hemorrhagic complications.

What do existing guidelines say?

There is no universal or specific guideline that directly addresses the interruption of antiplatelet therapy prior to graft biopsy in KT recipients.

Clinical practice points

- Consider patient-specific factors, such as age, platelet count, graft function, blood pressure and indication for antithrombotic therapy when deciding on the necessity of interrupting antiplatelet therapy.
- Low-dose ASA (≤150 mg/day) generally does not increase bleeding risk if discontinued at least 3–7 days before biopsy; continued ASA use within 3 days prior to biopsy may be associated with a higher risk of bleeding; discontinuing high-dose ASA (>300 mg) at least 5–7 days prior to biopsy should be considered to avoid bleeding.
- Discontinuation of clopidogrel at least 7 days prior to biopsy could be considered.

LIMITATIONS OF INCLUDED STUDIES

The studies included in our review have several important limitations. Firstly, most of the available evidence derives from retrospective single-center cohort studies or small case series, with only a handful of relevant prospective or randomized studies. This substantially increases the risk of selection bias and residual confounding. In many cases, the choice of antithrombotic therapy was at the discretion of the treating clinician, introducing potential indication bias. In addition, sample sizes were generally small or moderate, and outcomes of interest are relatively infrequent; as a result, many effect estimates are imprecise with wide confidence intervals, with several analyses underpowered to detect significant differences. Although multivariable models or propensity methods could have helped adjust for confounding, this was performed only by the minority of studies and even then, adjustment was often limited (e.g. incomplete data on cardiovascular risk profile, prior bleeding, concomitant antiplatelet use, immunosuppressive regimens or center-specific surgical practices).

Additionally, there was substantial heterogeneity across studies with populations varying widely with regards to type of transplant (living vs deceased donor), baseline thrombotic/bleeding risk, comorbidities, graft function and indication for antithrombotic therapy (e.g. AF, VTE, stent, primary prophylaxis, etc.). Interventions also differed significantly between studies in terms of drug classes, doses, timing of discontinuation/restart, bridging approach and concomitant therapies. This heterogeneity limits comparability and precludes definitive conclusions.

Outcome definitions and reporting practices were equally inconsistent. Bleeding and thrombotic events were not uniformly defined and follow-up windows ranged from immediate in-hospital post-operative period to several months. Some studies reported only transfusion requirements, others only reoperation rates, and many did not distinguish between minor and major bleeding. Perioperative INR levels, platelet counts, biopsy technique and use of reversal agents were often missing.

Taken together, these limitations result in the overall low certainty of evidence and high risk of bias of these studies. Therefore, our conclusions are framed as clinical practice points rather than strong, graded, guideline recommendations.

SUGGESTIONS FOR FUTURE RESEARCH

Rigorous, large-scale studies with standardized treatment protocols are needed to clarify the effects of pre-transplant antithrombotic therapy on transplant outcomes. Specifically, research should focus on the use of point-of-care platelet function tests to assess actual antiplatelet activity prior to deceased donor KT surgery. Pre-operative or intraoperative coagulation measured by thromboelastography can offer real-time insight into global hemostasis and help guide transfusion and perioperative antithrombotic decisions [108]. Given the increasing use of DOACs, especially apixaban, in the ESKD population, bleeding risk stratification tools and the role of DOAC reversal agents in KT candidates warrant further investigation. Additionally, large, prospective or propensity score–matched cohort studies comparing DOACs and warfarin in KT recipients could help determine optimal dosing and safety, and their impact on graft and patient outcomes. Research should also address patient-specific factors influencing bleeding risk in KT recipients on DOACs to improve risk stratification. Data on the use of betrixaban, a new DOAC, in CKD, and particularly in KT recipients, is currently lacking. Given its low renal excretion, this agent may offer a preferable profile [109]. Finally, FXI inhibitors, due to their potential to provide safe anticoagulation irrespective to kidney function, represent a promising research avenue in both transplant candidates and recipients [110, 111].

SUPPLEMENTARY DATA

Supplementary data are available at *Nephrology Dialysis Transplantation* online.

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DATA AVAILABILITY STATEMENT

No new data were generated or analyzed in support of this research.

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CONFLICT OF INTEREST STATEMENT

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